

PERCEPTION-INSPIRED BAYESIAN CAUSAL FUSION FOR AUDIOVISUAL SOURCE LOCALIZATION

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ABSTRACT

Multimodal fusion promises more accurate perception but only when the modalities share a common cause. When they do not, the second modality carries no information about the target, and fusing it can only corrupt the estimate. We cast this *whether-to-fuse* decision as Bayesian causal inference, following the optimal-observer model of human multisensory perception, and implement it as a plug-and-play layer on top of frozen audio and visual models for sound event localization and detection. The model infers a common-cause posterior over visible candidates, then gates precision-weighted fusion accordingly. Fusing unconditionally more than doubles the direction error, whereas the causal gate improves on-screen localization while limiting off-screen degradation, without any joint network retraining.

Index Terms—Audiovisual fusion, Bayesian causal inference, common-cause inference, multisensory cue combination, sound event localization and detection

1. INTRODUCTION

Humans combine sensory cues into a coherent understanding of the world, motivating a growing body of multimodal machine learning research [1, 2]. Multimodal learning outperforms unimodal learning when the modalities carry complementary information about a shared latent state [3]. Therefore, whether this fusion helps or harms depends on whether such a shared state exists.

Consider vision and hearing. Sound propagates around obstacles and arrives from every direction, but vision covers only a limited field of view. This asymmetry is natural for both humans and machines; a robot, a wearable device, or an AR headset likewise pairs a forward-facing camera with multi-channel microphones. A source that is occluded, outside the field of view, or behind the observer therefore remains audible but invisible (Figure 1). When a source *is* visible, vision offers higher spatial precision than audio [4, 5], so combining them reduces error. When it is not visible, the visual channel is conditionally independent of the source, and fusing it pulls the estimate toward an unrelated entity [6, 7].

Before asking *how* to fuse, the observer must first determine *whether* to fuse at all. Yet jointly trained networks delegate this decision to learned representations [8, 9] with no explicit gate. Uncertainty-aware methods weigh modalities by estimated reliability [10, 11, 12] but modulate influence continuously rather than inferring whether a shared cause exists. On-screen sound separation systems address a related trade-off by learning to suppress off-screen audio through calibrated audiovisual correspondence [7, 13], and negative-audio evaluations expose the failure mode when this distinction is ignored [6]. These methods require end-to-end train-

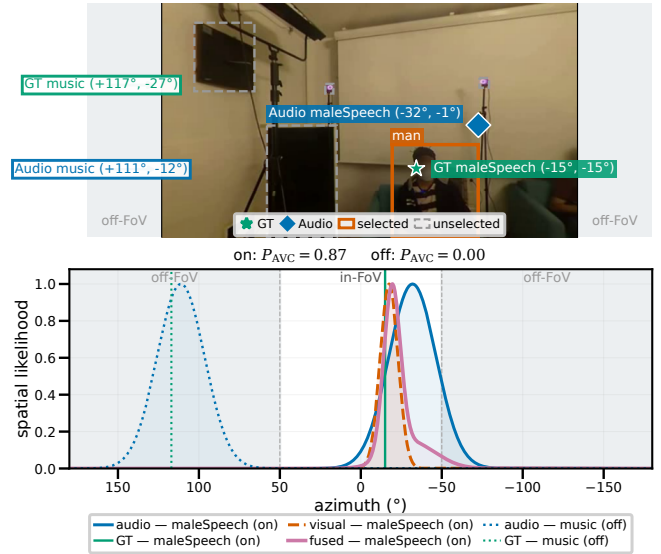


Fig. 1: Overview of Bayesian causal fusion on an example frame. Top: video frame with detected objects (orange: selected candidate; grey: unselected), ground-truth source location, and audio-only estimate for two concurrent sound events—an on-screen event (maleSpeech) with a compatible visible candidate and an off-screen event (music) with no matching candidate. Bottom: audio and visual angular Gaussians for each modality and the resulting precision-weighted fused estimate. For the candidate identified as sharing a common cause, the fused estimate is closer to the ground-truth location than the audio-only estimate.

ing on mixed on/off-screen data and target source separation or localization heatmaps rather than spatial event estimation.

Perception science offers a normative answer. Körding et al. [4] showed that human multisensory cue combination is well described by an *optimal observer* that performs *causal inference*: the brain first infers whether cues share a common cause, integrating them via precision-weighted combination when they do and segregating when they do not. The ventriloquist effect and its collapse at large disparities are natural consequences of this framework [4, 5, 14]. The entire benefit of integration depends on a shared cause, *i.e.*, without one, the second modality carries no information and integrating it can only degrade the estimate.

We target a complementary setting: post-hoc gating of frozen single-modality estimators for sound event localization and de-

tection (SELD) [15]. We call this approach *Bayesian causal fusion* (CAUSAL), operationalizing Körding et al.’s causal-inference observer [4] with three extensions: (i) generalizing the two-cue azimuth model to multiple visual candidate rays on the sphere, (ii) operating post-hoc on frozen audio and visual models—no component is retrained, and (iii) evaluating the resulting gate both as a common-cause classifier and as a localization corrector under DCASE on/off-screen protocols. We evaluate on the DCASE2025 Task 3 stereo on/off-screen dataset [16] as well as a four-channel microphone version [17] that we construct for full-sphere assessment. Frozen audio [16, 18] and visual [19] models serve as noisy estimators, much like biological sensors, in a limited-FOV (field of view) camera paired with spatial audio where the causal question arises naturally at every frame.

The remainder of this paper is organized as follows. Section 2 formulates the whether-to-fuse decision as Bayesian causal inference and derives the proposed multi-candidate fusion rule. Section 3 evaluates the approach on two DCASE datasets in terms of common-cause classification and localization accuracy. Section 4 concludes the paper.

2. METHOD

This section derives the Bayesian causal fusion model. We review the optimal-observer framework, define the audio and visual observations on the sphere, and present the proposed multi-candidate posterior and model-averaged integration rule in Sec. 2.4.

2.1. Optimal observer model

Following Körding et al. [4], consider an observer that must localize a source at true direction $\mathbf{s} \in \mathbb{S}^2$ from two noisy sensory measurements: an audio observation $\mathbf{x}_a$ and a visual observation $\mathbf{x}_v$. A latent causal variable C determines the generative structure (Figure 2):

- **Common cause** ($C = 1$): a single source at direction $\mathbf{s}$ generates both observations $\mathbf{x}_a$ and $\mathbf{x}_v$, which are then noisy measurements of the same $\mathbf{s}$, so integrating them reduces localization uncertainty.
- **Independent causes** ($C = 0$): the audio and visual signals originate from different sources, *i.e.*, $\mathbf{x}_v \perp \mathbf{s} \mid C=0$. The visual observation carries no information about the audio source direction, and the observer should segregate.

The optimal Bayesian observer must simultaneously infer the causal structure and estimate source location. Under squared error, the optimal estimate decomposes via the law of total expectation [4]:

$$\hat{\mathbf{s}}^* = p \hat{\mathbf{s}}_{\text{int}} + (1 - p) \hat{\mathbf{s}}_{\text{seg}}, \quad (1)$$

where $p = P(C=1 \mid \mathbf{x}_a, \mathbf{x}_v)$ is the common-cause posterior, $\hat{\mathbf{s}}_{\text{int}}$ is the precision-weighted integrated estimate under assumed common cause, and $\hat{\mathbf{s}}_{\text{seg}}$ is the segregated (audio-only) estimate under assumed independent causes. Full integration ($p=1$) and full segregation ($p=0$) are special cases, and the optimal observer interpolates between them weighted by its belief about causal structure. The common-cause posterior is the causal gate for multisensory integration, which determines whether visual data has any information value at all.

2.2. Audio and visual observations

We now instantiate the abstract framework with concrete direction estimates: the audio observation $\mathbf{x}_a$ becomes a decoded direction $\mathbf{a} \in \mathbb{S}^2$, and each candidate visual observation becomes a camera

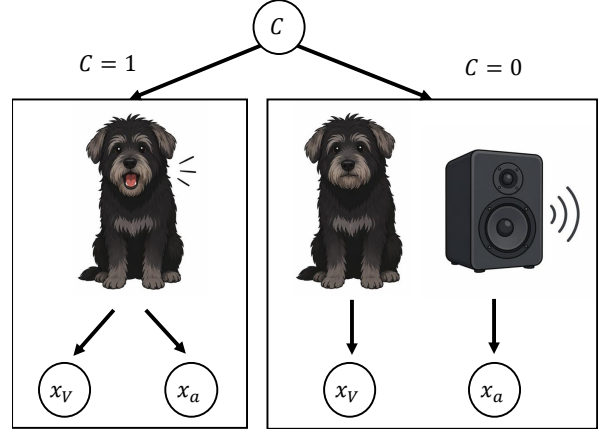


Fig. 2: The two causal structures. $C=1$: a single source (the dog) generates both the visual observation $\mathbf{x}_v$ and the audio observation $\mathbf{x}_a$, so integration is beneficial. $C=0$: the visual and audio signals have independent causes, so integration corrupts the estimate.

ray $\mathbf{v}_j \in \mathbb{S}^2$ indexed by detected object j . The binary causal variable generalizes to multiple visual candidates, $C \in \{0, 1, \dots, J_c\}$, where $C=j$ for $j \geq 1$ means that candidate j and the sound share a common source, and $C=0$ means that none of the visible candidates explains the sound, either because the source is off-screen, occluded, or missed by the detector.

At each 100-ms frame, a frozen audio model operating produces an event class c and a Cartesian unit direction $\mathbf{a} \in \mathbb{S}^2$. A frozen open-vocabulary detector is prompted with labels of potentially sounding objects (*e.g.*, *dog*, *vacuum cleaner*, *person*) and produces labeled bounding boxes $\{b_j\}_{j=1}^J$, each with class label ℓ_j . Because the detector and the audio model use different taxonomies, a manual mapping pairs each of the 26 detector labels with a compatible DCASE class (*e.g.*, *vacuum cleaner* $\rightarrow$ *domestic sounds*); only detections that match the current audio event c are retained as candidates $\mathcal{J}_c = \{j : \ell_j \in \mathcal{S}(c)\}$. In principle, an LLM or audio-language embeddings such as CLAP [20] can advance this step [21].

Each bounding box center (u_j, r_j) is projected to a unit camera ray with focal length $f = (W/2) \cot(\Omega/2)$, where Ω is the horizontal field of view:

$$\mathbf{v}_j = \frac{[f, W/2 - u_j, H/2 - r_j]^\top}{\|[f, W/2 - u_j, H/2 - r_j]^\top\|_2} \in \mathbb{S}^2, \quad (2)$$

where $W/2 - u_j$ and $H/2 - r_j$ are the pixel offsets from the image center along the azimuth and elevation axes, respectively.

2.3. Angular likelihood

Audio and visual localization uncertainty is estimated on training events as great-circle RMS errors σ_a and σ_v . Their common-cause separation has RMS $\sigma_\Delta = (\sigma_a^2 + \sigma_v^2)^{1/2}$, expressed in radians below. We model a candidate direction under a common cause by a von Mises–Fisher (vMF) density centered on $\mathbf{a}$,

$$p(\mathbf{v}_j \mid \mathbf{a}, C = j) = \frac{\kappa \exp(\kappa \mathbf{a}^\top \mathbf{v}_j)}{4\pi \sinh \kappa}, \quad \kappa = \frac{2}{\sigma_\Delta^2}. \quad (3)$$

Under $C=0$ (no common cause), a candidate direction is uniform over solid angle, $p(\mathbf{v}_j \mid C = 0) = 1/(4\pi)$. The spherical angular

Bayes factor is therefore

$$\Lambda_j = \frac{\kappa}{\sinh \kappa} \exp(\kappa \mathbf{a}^\top \mathbf{v}_j). \quad (4)$$

2.4. Causal posterior and fusion

Let $J_c = |\mathcal{J}_c|$ and let π_{vis} denote the training frequency of on-screen events among candidate-eligible samples, used as a proxy prior for a visible cause. This prior is divided equally among candidates: $P(C=j) = \pi_{\text{vis}}/J_c$, $P(C=0) = 1 - \pi_{\text{vis}}$. The common-cause posterior is

$$P(C=j \mid \mathbf{a}, \mathbf{v}_{1:J}, c) = \frac{(\pi_{\text{vis}}/J_c)\Lambda_j}{(1 - \pi_{\text{vis}}) + \sum_{k \in \mathcal{J}_c} (\pi_{\text{vis}}/J_c)\Lambda_k}. \quad (5)$$

For brevity, let $p_j \triangleq P(C=j \mid \mathbf{a}, \mathbf{v}_{1:J}, c)$ denote this posterior. We use $P_{\text{AVC}} = 1 - p_0$ as the common-cause posterior, the multi-candidate generalization of p in (1). When $J_c=0$, the independent-cause probability is 1.

Implementing the causal gate of (1), we first infer causal structure, then integrate. Conditioned on $C=j$, the precision-weighted integrated direction estimate is

$$\hat{\mathbf{s}}_j = \frac{\sigma_a^{-2} \mathbf{a} + \sigma_v^{-2} \mathbf{v}_j}{\|\sigma_a^{-2} \mathbf{a} + \sigma_v^{-2} \mathbf{v}_j\|_2}. \quad (6)$$

We propose to obtain the final estimate by directly implementing (1) as a multi-candidate model average on the sphere:

$$\hat{\mathbf{s}} = \frac{p_0 \mathbf{a} + \sum_{j \in \mathcal{J}_c} p_j \hat{\mathbf{s}}_j}{\|p_0 \mathbf{a} + \sum_{j \in \mathcal{J}_c} p_j \hat{\mathbf{s}}_j\|_2}. \quad (7)$$

The denominator projects the weighted combination back to the unit sphere. When $P_{\text{AVC}} \approx 0$, the visual terms vanish and the estimate reduces to the audio direction $\mathbf{a}$; when $P_{\text{AVC}} \approx 1$, it reduces to precision-weighted cue combination with the dominant candidate. Averaging over all candidates rather than selecting the most probable one ($\arg \max_j p_j$) avoids discontinuities: when two candidates carry similar posterior mass, a small change in the audio observation can flip the winner, causing the fused estimate to jump. The model average moves continuously as posterior mass shifts between candidates. No threshold is needed. The posterior continuously gates how much vision influences the estimate, reflecting the conditional independence $\mathbf{x}_v \perp \mathbf{s} \mid C=0$.

3. EXPERIMENTS AND RESULTS

We evaluate on two dataset configurations, assessing common-cause classification (Sec. 3.3), localization accuracy (Sec. 3.4), and comparison with the jointly trained DCASE2025 baseline (Sec. 3.5).

3.1. Setup

Datasets. We evaluate on two versions of the DCASE2025 Task 3 dataset [16], both derived from STARSS23 [17]. (i) *Stereo*: the original DCASE2025 release with stereo audio (azimuth only) and perspective video at 640×360 with horizontal FOV $\Omega=100^\circ$ [16]. (ii) *Four-channel microphone (MIC)*: a tetrahedral array counterpart using the same temporal crops and camera yaws, providing full-sphere (azimuth + elevation) evaluation. The microphone directions are rotated to the camera viewing direction so that audio and visual rays share the same coordinate frame. Multi-channel arrays are

Table 1: Common-cause classification on eligible events using two proxy labels. Higher AUROC and AP indicate better discrimination; lower Brier score and ECE indicate better calibration. Calibration metrics are reported only for the probabilistic CAUSAL posterior.

Method	Proxy	AUROC	AP	Brier	ECE
FORCED	on-screen	0.500	0.332	—	—
GATE-20	on-screen	0.828	0.723	—	—
CAUSAL (proposed)	on-screen	0.920	0.847	0.093	0.060
FORCED	common-cause	0.500	0.302	—	—
GATE-20	common-cause	0.839	0.706	—	—
CAUSAL (proposed)	common-cause	0.924	0.805	0.085	0.048

standard for spatial scene understanding [17], making this a more realistic setting.

No ground-truth common-cause label exists, so we evaluate the common-cause posterior against two proxies. (a) *On-screen*: the DCASE2025 human annotation of whether the source is within the field of view [16]. This is a necessary precondition for a visual common cause, but permissive: a source may be on-screen without a matching detection. (b) *Common-cause*: an on-screen event whose ground-truth location lies within 20° of a semantically compatible Grounding DINO detection. This is a tighter approximation of whether a matched visual candidate exists, but it shares the detector and semantic mapping with the causal gate’s candidate set, so high agreement is partly a consequence of shared inputs rather than independent validation of causal inference.

Models. For stereo, we use the official DCASE2025 checkpoints¹: audio-only SELDnet (AUDIO) and jointly trained audio-visual SELDnet (AV-BASE). For microphone, we use an available open source model from the DCASE2024 challenge [18]². In both settings, Grounding DINO Tiny [19] serves as the open-vocabulary detector, prompted with a fixed vocabulary of 26 sounding-object labels. The semantic mapping $\mathcal{S}(c)$ (Sec. 2) pairs each detector label with one of the 13 DCASE classes. All components are frozen, and causal fusion is applied post-hoc as described in Sec. 2.

3.2. Compared methods

Audio and visual detections are fixed across all methods; only the fusion strategy varies. We compare four strategies that differ in how/whether the common-cause posterior P_{AVC} gates fusion, with the audio-only estimate being the baseline and the last being the method from Sec. 2.4:

- AUDIO: retain audio direction ($P_{\text{AVC}}=0$; full segregation).
- FORCED: always fuse with the nearest compatible candidate ($P_{\text{AVC}}=1$; full integration).
- GATE-20: fuse only when audio-visual disparity $< 20^\circ$ (hard angular gate).
- CAUSAL (Bayesian causal fusion; the proposed method): infer P_{AVC} via (5), then model-average via (7).

AUDIO and FORCED are the two limiting cases of the causal inference model [4]. GATE-20 is an ad-hoc threshold. CAUSAL tests whether the normative solution provides a better gate.

3.3. Common-cause classification

Before evaluating localization, we ask whether the inferred common-cause posterior discriminates events for which visual integration is

¹https://github.com/partha2409/DCASE2025_seld_baseline

²<https://github.com/axeber01/ngcc-seld>

Table 2: MIC full-sphere class-macro DOAE ($^{\circ}$) on eligible events, following the official DCASE protocol. Δ ($^{\circ}$) is relative to AUDIO. Lower is better.

Method	DOAE ($^{\circ}$)			Δ ($^{\circ}$) from audio	
	All	On	Off	On	Off
AUDIO	30.41	28.15	30.95	0.00	0.00
FORCED	61.47	12.74	89.20	-15.41	+58.26
GATE-20	29.64	26.21	31.03	-1.94	+0.08
CAUSAL (proposed)	29.50	25.40	31.20	-2.75	+0.26

plausible (Table 1), using the two proxies from Sec. 3.1.

An event is *eligible* when the detector finds at least one compatible object; otherwise all methods reduce to the audio-only estimate. Of the eligible events, only 33.2% are on-screen, while 66.8% are off-screen. The stricter proxy identifies 30.2% as plausible common-cause cases.

The detector-independent on-screen label provides the stronger test. The continuous posterior discriminates on-screen from off-screen events (AUROC 0.920, AP 0.847; Table 1), substantially exceeding chance. Because GATE-20 is a binary decision, its AUROC reduces to balanced accuracy at a single operating point; the localization comparison in Sec. 3.4 provides the primary evaluation. Against the common-cause proxy, CAUSAL achieves AUROC 0.924 and AP 0.805. This second result is less independent because the proxy reuses the detector and semantic mapping that construct the model’s candidate set. Neither proxy directly annotates causal structure, so these results demonstrate the gate’s discriminative value rather than recovery of a latent causal variable.

Because the posterior is used as a continuous fusion weight, discrimination alone is insufficient: its probabilities must also be calibrated. We measure calibration with the Brier score (mean squared error between predicted probability and label) and expected calibration error (ECE; mean absolute gap between predicted confidence and observed accuracy across probability bins). CAUSAL obtains a Brier score of 0.093 and an ECE of 0.060 against the on-screen proxy. Against the common-cause proxy, the corresponding values are 0.085 and 0.048. Thus, within a semantically compatible candidate set, spatial disparity alone yields a discriminative and well-calibrated gate with respect to these proxies, without spectral, temporal, or learned cross-modal features.

For reproducibility, training-split estimation yielded $\sigma_a=14.7^{\circ}$, $\sigma_v=5.5^{\circ}$, and $\pi_{vis}=0.33$; all three values were frozen before test-set evaluation.

3.4. Localization

Table 2 reports the effect on localization. Since only a minority of eligible events are on-screen, overall gains are modest even when per-condition effects are large. FORCED fusion improves on-screen DOAE but roughly doubles eligible-event error, because off-screen events are pulled toward unrelated detections. This is the cost of integrating when $C=0$.

Table 2 also reveals that CAUSAL improves on-screen DOAE by 2.75° while limiting off-screen degradation to $+0.26^{\circ}$, reducing eligible-event error by 0.91° . Its on-screen gain exceeds that of GATE-20 (2.75° vs. 1.94°) because the posterior weights strong candidates more heavily instead of applying one fixed threshold.

Table 3: DCASE2025 stereo comparison on the test split (class-macro DOAE, following the official protocol). AUDIO and AV-BASE are the two official DCASE2025 baseline checkpoints [16]; CAUSAL applies post-hoc causal fusion to AUDIO’s detections. DOAE ($^{\circ}$, lower is better); on/off-screen accuracy (higher is better).

Method	DOAE ($^{\circ}$)			On/Off
	All	On	Off	Accuracy
AUDIO	24.58	16.59	25.74	—
AV-BASE	22.16	19.13	22.89	0.74
CAUSAL (proposed)	24.24	13.78	25.96	0.76

3.5. Comparison with DCASE 2025 Task 3 baseline

The DCASE 2025 Task 3 baseline includes a jointly trained audio-visual model, enabling a direct comparison with a learned fusion approach. We apply causal fusion to the stereo dataset using the official audio-only checkpoint [16]. Since stereo front-back ambiguity folds azimuths into $[-90^{\circ}, 90^{\circ}]$, the vMF likelihood reduces to a wrapped Gaussian; the model is otherwise identical to Sec. 2. We compare against both official baselines (AUDIO and the jointly trained AV-BASE [16]). On/off-screen classification accuracy is defined as $\mathbf{1}[P_{AVC} > 0.5]$ for CAUSAL and the model’s internal flag for the AV-BASE.

Table 3 shows that CAUSAL reduces on-screen DOAE by 2.81° relative to AUDIO. The jointly trained AV-BASE achieves the lowest aggregate and off-screen DOAE, but its on-screen DOAE (19.13°) is worse than both AUDIO (16.59°) and CAUSAL (13.78°), indicating that joint training does not learn an effective on/off-screen gate. CAUSAL attains higher on/off-screen accuracy (0.76 vs. 0.74) without any joint training.

4. CONCLUSION

We cast the *whether-to-fuse* decision as Bayesian causal inference, following the optimal-observer model of human multisensory perception [4], and applied it as a plug-and-play layer above frozen audio and visual models for sound event localization and detection. Without the gate, fusion with off-screen sources drastically raises direction error, because the visual channel is conditionally independent of the source. With the gate, precision-weighted cue combination captures the on-screen benefit while containing off-screen harm. Because the gate operates post-hoc on frozen models, it is plug-and-play: no multimodal training is needed, and any improved audio or visual model can be substituted directly. On two datasets, this zero-training-cost fusion improves on-screen DOAE over the audio-only baseline. On the stereo dataset, it also achieves better on-screen localization and on/off-screen classification than a jointly trained audiovisual system [16]. The gate currently relies on an open-vocabulary detector and a manual binary class mapping; failures in either reduce the candidate set and default to audio-only, which is safe but forgoes visual benefit. A natural extension is to replace the hard semantic filter with a soft compatibility score from an audio-language model such as CLAP [20], removing the need for a manual class mapping. The same principle of inferring whether modalities share a common cause before combining them may apply to other multimodal settings where the second modality is not always informative.

5. ACKNOWLEDGMENTS

This work was supported by the HUCE infrastructure of the Aalto School of Electrical Engineering. SJS were supported through the joint German Academic Exchange Service (DAAD) and Research Council of Finland (RCF) Project (57763119) "Immersive Augmented Acoustics (IAA)".

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